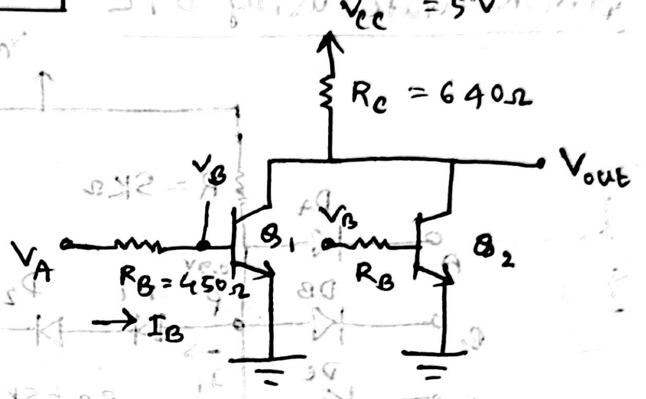
EEE 4483

NOR Gate using RTL

Truth table :

V_A	V_B	V_{out}
0	0	1
0	1	0
1	0	0
1	1	0



LOW → 0V
HIGH → 5V } Input

$\boxed{\text{Case 1}}$: 0, 0

$I_B = 0$, both transistors are in cutoff. V_{cc} fully goes to V_{out}.

So, $V_{out} = 5V$ ($I_C = 0$)

$\boxed{\text{Case 2}}$ $V_A = 5V$; $V_B = 0$

Q_2 cutoff.

Q_1 satn → 0.2 V

$V_{out} = 0.2V$ (logic 0)

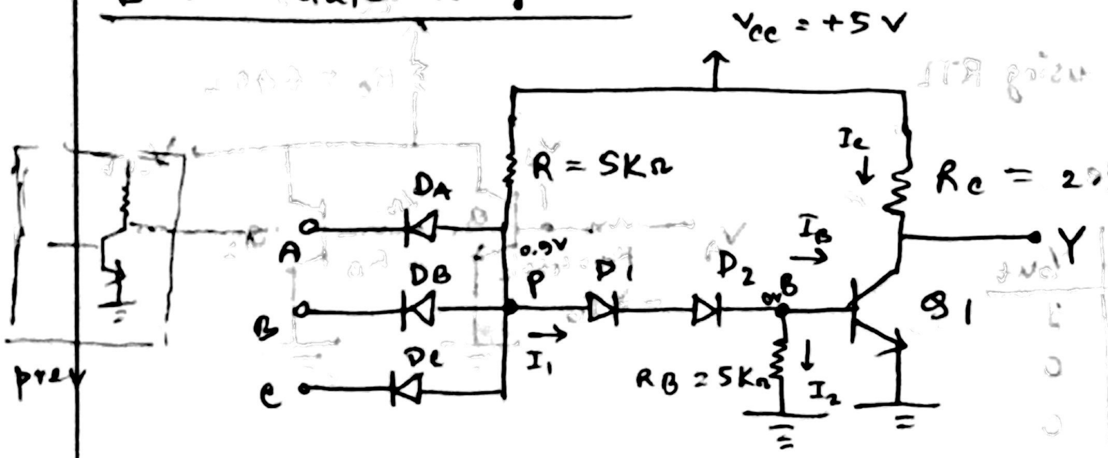
$\boxed{\text{Case 3}}$: 5V, 5V

Both transistors sat^n.

Both gives 0.2 V

same voltage parameters & noise margin as NOT Gate

NAND Gate using DTL



logic low → 0.2V

logic High → 5V

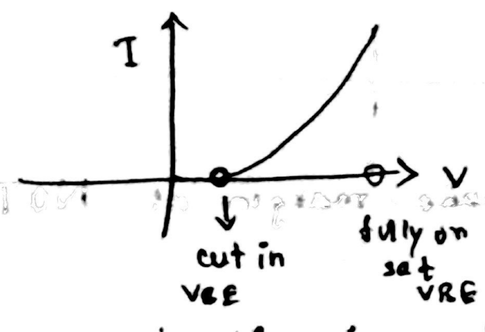
Assumptions

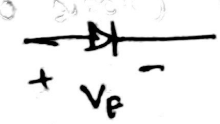
$V_{BE}(sat) = 0.8V$ (not same as active, was 0.7V in active)

V_{BE} (cut in) = 0.5V (BE Junction just starts to conducts, what is the voltage required)

V_{CE} (sat) = 0.2V

Voltage drop across a conducting diode = 0.7V





* BE junction is a diode *

Truth table

Truth Table

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

approved by case 1

$\boxed{\text{Case 1}}$

$V_A = 0.2V$ Low
$V_B = 5V$ HIGH
$V_C = 5V$ HIGH

- for logic Low, gate transistor is saturated and output is 0.2V
- for logic HIGH, gate transistor is turned off and output voltage is 5V

$D_B, D_C \rightarrow$ reverse biased

$D_A \Rightarrow$ forward biased

$V_P = ?$

A is Low state, its diode D_A conducts

$\therefore V_P = 0.2 + 0.7 = 0.9V$

(from input A) (D_A voltage drop)

Now, if D_1 and D_2 path conducts,
both D_1 and D_2 has to be frward biased.

But, 0.9V is not enough to turn both the diodes ON.

So, Q_1 is turned off.

$\therefore V_{out} = V_{cc} = +5V$

Whenever any of the input is LOW, output will be HIGH

Case 2

$V_A = 5V$

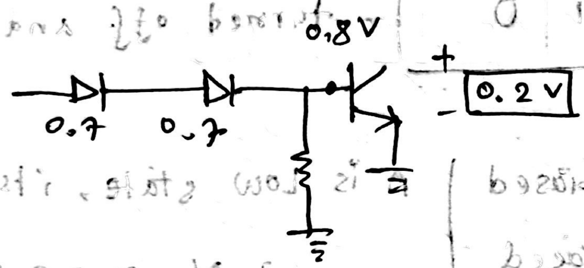
$V_B = 5V$

$V_C = 5V$

We need at least 5.7 V in anode side for the diodes to be turned on.

So, $D_A, D_B, D_C \rightarrow$ all reverse biased. OFF

sat^n.



$\therefore 0.7 + 0.7 + 0.8\ V = 2.2\ V$

$V_p = 2.2V$

$\therefore V_Y = 0.2V$ (LOW)

Values of I_1, I_2, I_B, I_C :

$$I_1 = \frac{V_{cc} - V_P}{R} = \frac{5 - 2.2}{5K} = 0.56\ mA$$

$$I_2 = \frac{V_{BE}(sat)}{R_B} = \frac{0.8}{5K} = 0.16\ mA$$

$$I_B = I_1 - I_2 = (0.56 - 0.16)\ mA = 0.4\ mA$$

$$I_c = \frac{V_{cc} - V_{CE(sat)}}{R_c} = \frac{5-0.2}{2.2K} = 2.18\ mA$$

Purpose of D_2 :

In case 2, D2 not required.

But in case 1, V_p was 0.9V.

So, D1 would be turned ON. But, would not be enough to turn Q1 ON

$V_B = 0.2V$

Transistor would not reach sat^n

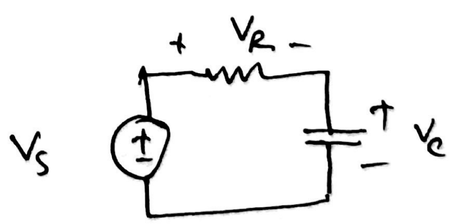
But, Diff noises can cause fluctuations.

V_p may change

and V_B can also change and change the Q_1 to sat^n.

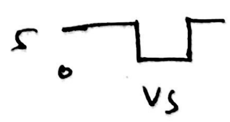
So, using two diodes is good design and safe for practical cases (even three are used practically)

Purpose of R_B :



Theoretically, fine.

Stores charge in base.



Supply volt changes abruptly.

If measuring V_R, without delay it changes with V_s. But V_c takes time.

$\boxed{\text{Storage time}}$ ↑ Delay of gate ↑

To decrease storage time, R_B should be decreased.

↳ V_e

Similarly, for transistor when mode changes, charge stored in base

If Inputs changed, $V_p = 2.2V$ to $0.9\ V$,

cannot change abruptly.

$0.9V \longleftrightarrow 0.8V$

0.1 V → not enough to turn two diodes ON.

But the charge stored must be dissipated somewhere.

Diode path is open ckt now. So, this charge goes through R_B.

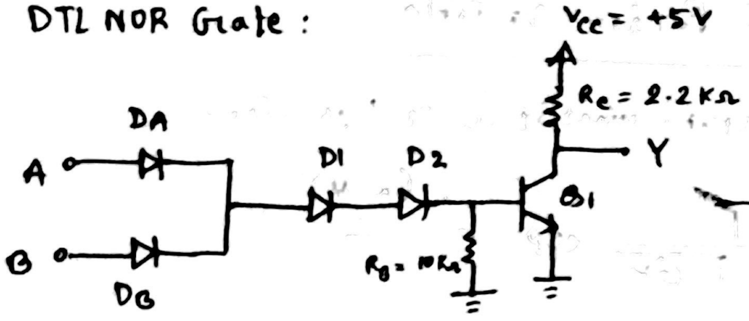
We need higher R_B → so quickly dissipates

" " lower R_B → so I_B increases (when all inputs ON, more current goes through R_B)

→ tradeoff made (somewhere in the middle)

DTL NOR Gate :

HIGH - 5V
LOW - 0.2V



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

$0.7+0.7+0.7+0.8 = 2.9V$

one input NAND Gate → becomes NOT Gate

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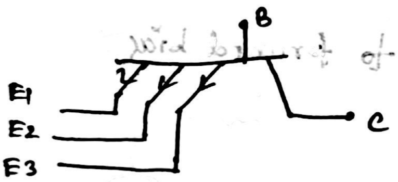
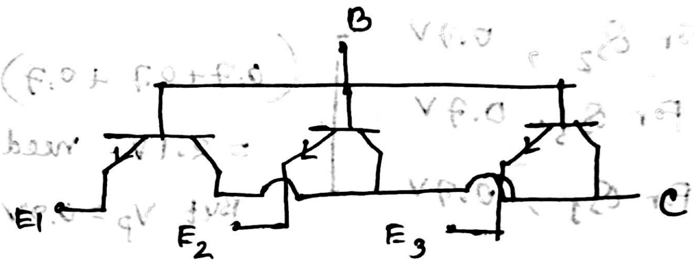
Transistor–Transistor Logic (TTL)

Multi-emitter transistor





only collector and base pins connected with each other

$B = 0.7V$ Because of E_3, forward biased.

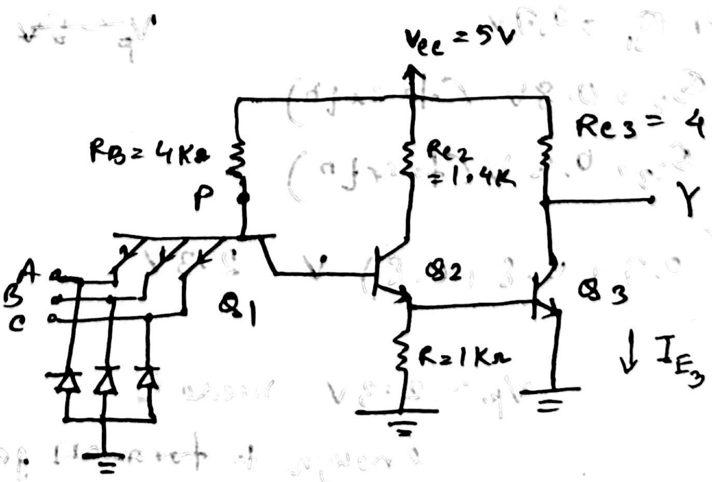
$E1 = 5V$ at least one of the Emitters less than 0.7 (B)

$E2 = 5V$ means entire thing turned on.

$E3 = 0V$

Reverse active mode : CB forward, BE reverse

TTL NAND Gate



(Remove diodes from fig)

$\boxed{\text{case 1}}$

at least one LOW

A = 0.2V
B = 5V
C = 5V

$V_p = (0.2 + 0.7)\ V$
$= 0.9V$

A low means that junction turned ON. Forward biased.

For Q_2, 0.7V
For Q_3, 0.7V
For Q_1, 0.7V

$(0.7 + 0.7 + 0.7)\ V$
$= 2.1V$ needed to forward bias
But $V_p = 0.9V$

$\therefore$ Reverse Bias,
Rest ckt → open ckt
$\therefore V_o = V_{cc} = +5V$ (HIGH)

$\boxed{\text{Case 2}}$

All HIGH

A = 5V
B = 5V
C = 5V

Reverse biased

For Q_1, 0.7V

~~$V_p = 5V$~~

Q_2, 0.8V (to satⁿ)
Q_3, 0.8 V (to satⁿ)

$(0.7 + 0.8 + 0.8)\ V = 2.3V$

$V_p = 2.3V$ needed
enough to ~~turn on~~ get satⁿ
$V_o = 0.2V$

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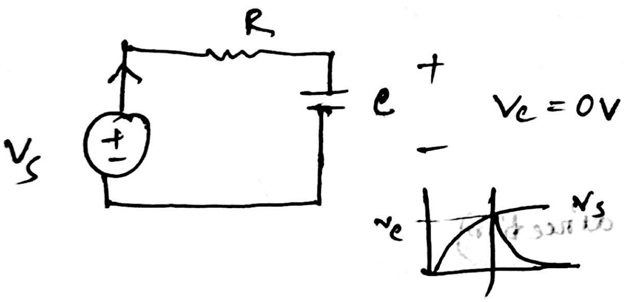
* When all inputs high,

Q_2 and Q_3 in sat^n $\Rightarrow$

Now, one input changed to Low,

Now, $V_P = 0.9V$ (from 2.3 V)

(active mode) due to reverse current, that stored charge can be dissipated. This is faster than that R_B in DTL NAND.



$\tau = RC$

if $RC \uparrow$ more time to get dissipated.

Purpose of three diodes:

due to noise, can get neg voltage at the input the diode helps to sth, idk

$\boxed{\text{Math}}$

In the ckt of TTL NAND Gate, if all the inputs are in HIGH condition, then determine the emitter current of Q_3

Soln : $V_A = 5V$

$V_B = 5V$

$V_C = 5V$

$Q_1 \rightarrow$ reverse active mode

- BE junction reverse biased
- CB junction forward biased

$$V_P = V_{BC1} + V_{BE2} + V_{BE3}$$

$$= 0.7 + 0.8 + 0.8$$

$$= 2.3V$$

$$I_{B1} = \frac{V_{CC} - V_P}{4K\Omega} = \frac{5 - 2.3}{4K} = 0.675mA$$

$$I_{E1} \approx 0$$

$$0 = I_{B1} + I_{C1}$$

$\Rightarrow I_{B1} = -I_{C1}$ (opposite direction)

$$= I_{B2}$$

$$= 0.675mA$$

$$V_{BE3} \approx V_{B3} = 0.8V$$

$$I_R / I_{1K\Omega} = \frac{V_{B3}}{1K\Omega} = \frac{0.8}{1K} = 0.8mA$$

$$V_{C2} = V_{B3} + V_{CE2} = 0.8 + 0.2 = 1V$$

$$I_{c2} = \frac{V_{cc} - V_{c2}}{1.4k} = \frac{5-1}{1.4K} = 2.86mA$$

$$I_{E2} = I_{B2} + I_{c2} = (0.675 + 2.86) = 3.53 mA$$

$$I_{B3} = I_{E2} - I_{1K} = 3.53 - 0.8 = 2.73 mA$$

$$I_{c3} = \frac{V_{cc} - 0.2}{4K} = \frac{5-0.2}{4K} = 1.2 mA$$

$$I_{E3} = I_{B3} + I_{c3} = 2.73 + 1.2$$

$$= 3.43 mA$$

EEE

4 TTL : Improvement in output stages

Load Capacitance : Total capacitance that the output of a logic gate must charge or discharge during a logic transistor. (HIGH ↔ LOW)

This capacitance introduces delays de in signal transitions, limiting switching speed.

Sources of Load capacitance

→ Every TTL input pin behaves like a small capacitor due to the base-emitter junction of the input BJT.

{ in reverse biased condition → capacitance due to depletion region acting as dielectric between two charge regions. }

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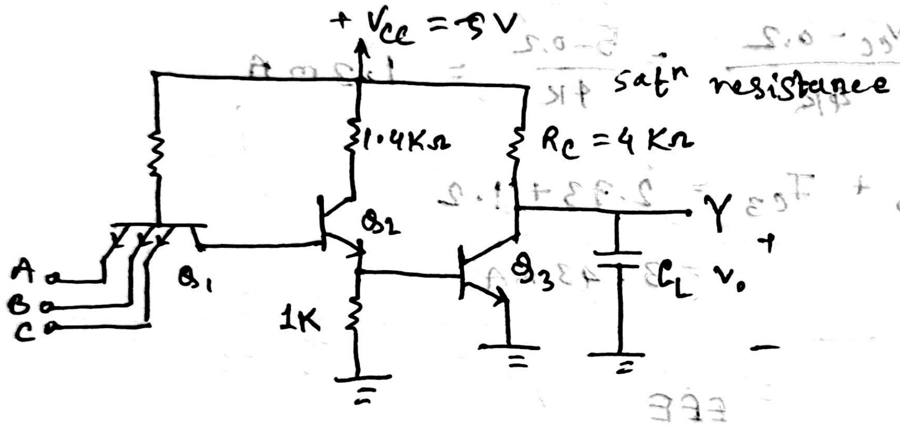
→ The physical wires or PCB traces connecting gates act like capacitors between trace to GND, trace / signal line to adjacent signal lines.



TTL output must charge and discharge this load capacitance every time the logic level changes.

If passive pull up resistor R_C is used for this purpose →

time constant $R_C C_L$ is high, causing long delay time when logic level experiences a transition.

* When $5 \rightarrow 0.2V$

effective resistance is low, time constant is low

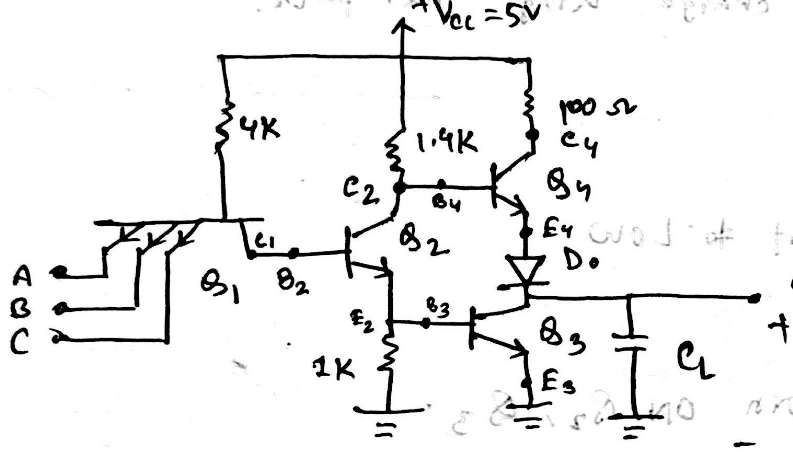
* But when $0.2 \rightarrow 5V$

effective resistance high, time constant high

- Why not just use low R_C?

→ problem in other cases — ian what

- Solution : Totem pole config.



Let, $Y = 0.2V$ (LOW condition)

when $V_A = V_B = V_C = 5V$

$Q_2 \to 0.8V$ (sat^n)

$Q_3 \to 0.8V$ (sat^n)

$V_{C2} = V_{CE2}(sat) + V_{BE3}(sat)$

$= 0.2V + 0.8V$

$= 1V$

$V_{B4} = V_{C2} = 1V$ [as B_4 and C_2 are same node

Ideally, we want Q_4 turned off. [If both turned ON, unnecessary current flow]

But is that happening?

→ if no diode D_0, $V_{C2} = V_{B4} = 1V$

$\therefore V_{E4} = 0.2V$ (since Q_3 sat^n)

$\therefore V_{BE4} = V_{B4} - V_{E4} = 0.8V$ (enough for sat^n)

This is why, diode is there.

Now, both diode and Q_4 have to be turned ON using 0.8V. Not possible so turned OFF.

Now, can dissipate charge using other path.

Transition

change one input to Low

$V_{B_1} = 0.9\ V$

not enough to turn ON Q_2, Q_3.

So, $Q_2, Q_3 \rightarrow$ cutoff

open ckt path

direct $V_{cc} = 5V$ comes to V_{B4}. Q_4 is saturated

output should be 5V. But takes time for capacitor to charge up to 5V.

How?

→ Q_4 Needs to be active/saturated

B_4 : 0.8 V	
Do : 0.7 V	V_{C2} 1.7 V needed.
prev: 0.2 V	supply - 5V : Yes, possible.

→ $V_{CE4} = 0.2\ V$ (sat^n)

V_{cc} $\xrightarrow{I_x}$ current flows in this path. capacitor gets charged up

Q_4

e_L

$$I_x = \frac{5 - 0.2 - 0.7 - 0.2}{100}$$

it keeps increasing, reducing I_x

$I_x = I_{e4}$

charging up will stop when no current

diode and Q_4 turned OFF

collector current reducing means transistor is going toward active from satn. Then at an equilibrium, both Q_4 and diode at a state betn ON and OFF, $V_o = 3.9V$.

$$V_o = V_{cc} - V_{BE4}\ (\text{cut in}) - V_{D0}\ (\text{cut in})$$

$$= 5 - 0.5 - 0.6$$

$$= 3.9V$$

$$\boxed{\begin{array}{l} V_{c2} = 5V \\ V_{c3} = 3.9V \end{array}}$$

sacrificed some less volt at output in HIGH in totem pole.

Purpose of 100Ω resistor:

at a time, Q_3, Q_4 should not turn ON.

But for small time, it can happen. if didnt have the resistor huge amount of current I_x.

Q_4 should turn ON earlier.